

Learning Computer Graphics via Authentic Engineering Challenges

Lei Yang

The University of Hong Kong
Hong Kong, China
yanglei.dalian@gmail.com

Abstract

Computer graphics is traditionally introduced to undergraduates through topics such as rendering, modeling, and animation, which can obscure its importance in engineering domains where spatial reconstruction and understanding are critical. This paper reports a case study from the Tam Wing Fan Innovation Wing at The University of Hong Kong, where computer graphics learning is embedded in an authentic engineering challenge: reconstructing coastal infrastructure from noisy LiDAR-SONAR point clouds. An interdisciplinary student team explored a staged sequence of methods—from statistical outlier removal and PCA-based point classification to Poisson surface reconstruction and neural implicit representations (SIRENs)—and achieved effective denoising performance in a real-world setting. We describe the accompanying pedagogical approach, emphasizing facilitative mentorship, technical scaffolding, and close collaboration with industrial partners. Our experience suggests that such authentic, project-based activities not only deepen students’ conceptual and technical understanding of computer graphics, but also broaden their perception of the field as an enabling technology for solving real engineering problems.

CCS Concepts

• **Computing methodologies** → **Shape modeling**; • **Applied computing** → **Education**.

Keywords

Experiential learning, 3D reconstruction

ACM Reference Format:

Lei Yang. 2026. Learning Computer Graphics via Authentic Engineering Challenges. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator’s Forum (SIGGRAPH Educator’s Forum ’26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3799829.3812526>

1 Introduction

Computer graphics is often associated with rendering, modeling, and animation, particularly in film-making and gaming. However, it also plays a crucial role in engineering applications such as surface design, physical simulation, manufacturing, and construction, where spatial reconstruction and understanding are essential.

Despite these rich connections to engineering practice, many undergraduate students remain unaware of them. Standard computer graphics curricula often emphasize the aforementioned core themes, which narrows students’ perception of the field and obscures its broader industrial and societal relevance. Broadening students’ understanding of what computer graphics can enable is therefore essential not only for nurturing future talent in the engineering sector, but also for making computer graphics a more visible and attractive area of study.

In this paper, we report recent efforts at the Tam Wing Fan Innovation Wing (InnoWing), a student active learning center under the Faculty of Engineering at The University of Hong Kong, to situate computer graphics learning within authentic engineering challenges. We describe one representative student project that leverage computer graphics technologies to address real-world problem: 3D reconstruction of coastal infrastructure from noisy point cloud scans produced by integrated LiDAR-SONAR sensors; see Figure 1. We then highlight how these projects were structured as learning activities and how they helped foster students’ interest and competency in computer graphics.

These efforts are enabled by the vision and philosophy of the Tam Wing Fan Innovation Wing at The University of Hong Kong, which provides an open, hands-on environment to foster interdisciplinary innovation and experiential learning among engineering undergraduates. InnoWing collaborates with academic and industrial partners to curate project-based learning opportunities where students apply technical skills to real-world, societal challenges [Chui and Tien 2024; YANG et al. 2025a,b].

2 Surface Reconstruction of Coastal Infrastructure from Noisy Point Cloud Scans

This project was initiated through a collaboration with the surveying unit of a local government office responsible for coastal infrastructure maintenance. Early discussions identified a key challenge: although surveyors can efficiently collect scans of coastal structures using ships equipped with integrated LiDAR-SONAR sensors, the resulting point clouds are often severely degraded by water surface reflections and underwater disturbances such as surges and aquatic vegetation. Denoising and preprocessing currently rely on manual cleaning, a labor-intensive process that can take several days per site.

Since the summer of 2025, an interdisciplinary team of four students, two from computer science and two from civil engineering, has been working on this problem. They explored two complementary approaches: (1) classifying individual points as either structural



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ACM ISBN 979-8-4007-2552-4/26/07
<https://doi.org/10.1145/3799829.3812526>

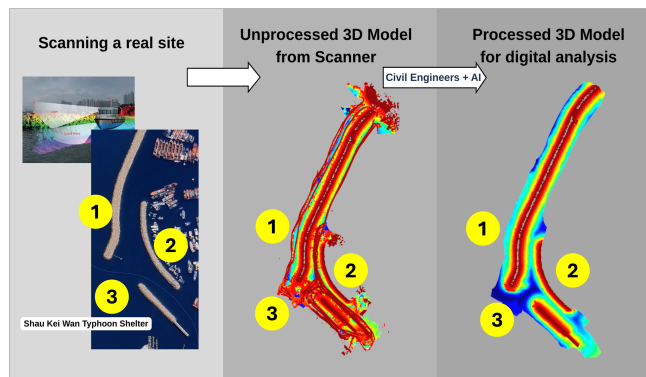


Figure 1: An overview of the project pipeline that reconstructs coastal structures from noisy point cloud scans. The color indicates the altitude of the points above the sea level. The middle subfigure shows the noisy scans from the process illustrated in the left. The rightmost subfigure shows a clean data of the coastal structure produced by the method developed by the student team.

features or noise artifacts, and (2) reconstructing smooth continuous surfaces to approximate the seabed and coastal structures using implicit geometric representations.

The students began by applying statistical outlier removal methods implemented in Open3D [Zhou et al. 2018], a modern open-source library for 3D data processing. They then implemented principal component analysis (PCA) to classify points based on local geometric properties, computing planarity metrics from the eigenvalues of the local covariance matrix to distinguish flat structural surfaces from random noise patterns. In parallel, they experimented with implicit surface reconstruction, starting from Screened Poisson Reconstruction [Kazhdan and Hoppe 2013] and progressing to SIRENs (Sinusoidal Representation Networks) [Sitzmann et al. 2020], a neural implicit representation capable of capturing fine geometric details while maintaining smoothness.

Through this progressive development, the students gained a solid understanding of both discrete point-based methods and continuous implicit surface fitting for denoising. They proposed a workflow that combines PCA-based outlier removal with a SIREN-based surface fit to the filtered point cloud, achieving an average F1 score of 0.82 across six coastal sites. The pipeline is shown in Figure 1.

Building on this foundation, they now plan to investigate deep learning-based point cloud feature extraction methods [Qi et al. 2017a,b; Wang et al. 2019] to further automate noisy point classification. The project addresses a genuine industrial need while exposing students to the full pipeline of applied computer graphics research, from problem formulation and literature review to implementation and validation with domain experts.

3 Pedagogical Approach and Lessons Learned

As an educational makerspace, the Tam Wing Fan Innovation Wing adopts an experiential learning philosophy [Kolb 2014] in its co-curricular activities. Faculty and staff work closely with industrial

and academic partners to identify authentic problems and translate them into student projects that emphasize real-world impact and societal relevance. In this way, InnoWing serves as a bridge between the university and external stakeholders, enabling students to tackle genuine engineering needs through technology-driven solutions.

Facilitative mentorship and technical scaffolding. Throughout the project development process, our teaching team adopts a facilitative mentorship philosophy [Kurti et al. 2014] rather than a directive one. Instead of providing predefined solutions, we offer students conceptual “tools” and technical scaffolding so that they are gradually equipped with the competencies needed to progressively advance their work.

When students began the surface reconstruction project described in Section 2, most were first-year undergraduates with little or no prior experience in 3D modeling or graphics programming. Although a PointNet-like deep learning approach initially appeared promising for point cloud denoising, we recognized that they needed more foundational exposure. We therefore first guided them to explore the Open3D library [Zhou et al. 2018], whose examples and tutorials on point cloud processing helped them quickly develop programming confidence and technical fluency. Building on this foundation, we introduced Screened Poisson Surface Reconstruction [Kazhdan and Hoppe 2013], allowing them to experiment with implicit surface reconstruction and compare its performance with simpler denoising methods; the staged contrast between these approaches prompted reflection on geometric principles and algorithmic trade-offs and motivated iterative refinement, even when early results were imperfect.

As their understanding grew, we revisited the mathematical foundations of implicit representations, helping students connect theoretical coursework to concrete graphics applications. With this grounding, they advanced to implementing neural implicit representations such as SIRENs [Sitzmann et al. 2020], encountering practical challenges in data normalization, sampling strategies, and parameter tuning, and strengthening their systematic debugging and problem-solving skills through trial and error. Equipped with this progression of experiences—from basic libraries to classical reconstruction methods to neural representations—the team is now able to propose its own next research steps, including a PointNet-like, data-driven approach to compare against their existing rule-based pre-filtering strategy for noisy point detection. This evolution from guided exploration to self-directed research exemplifies how staged scaffolding and authentic challenges can nurture independent, creative, and technically competent engineers.

Teamwork and collaboration with industrial professionals. Students also developed teamwork and communication skills by working in interdisciplinary teams and regularly interacting with industrial professionals. Civil engineering students contributed domain knowledge about coastal structures and survey practices, while computer science students focused on algorithms and implementation. Exchanges between students from different backgrounds reinforced civil engineering students’ programming skills regarding 3D modeling and deepened computer science students’ understanding of how computer graphics can tackle real-world engineering problems.

Table 1: Classification of the approaches experimented in the project described in Section 2. The approaches are numbered based on the order they were introduced to the students. A systematic understanding is developed among the students, partly evidenced by that the students combined the PCA with SIRENs to achieve satisfactory denoising performance (F1 score: 0.82), and partly supported by the fact that they suggested to test PointNet-like approaches in the future.

	Traditional	Learning-based
Point-wise classification	Statistical outlier removal (1) PCA (3)	PointNet-like approaches (to-do)
Implicit representations	Poisson Reconstruction (2)	SIRENs (3)

Meetings with practicing engineers gave students insight into manual workflows for point cloud cleaning and inspection, which in turn inspired them to design more efficient automated methods that mimic these processes using cross-sectional views and iterative scans. Presenting intermediate results to external partners trained students to communicate technical content to diverse audiences and to incorporate feedback into subsequent iterations.

Beyond technical discussions, these interactions helped students appreciate differing priorities and constraints on the industrial side, such as safety, cost, and maintainability. Within the team, students learned to negotiate design choices, resolve disagreements constructively, and coordinate milestones around external partner timelines. Taken together, authentic problems, facilitative mentorship, and close industry engagement helped students experience computer graphics not only as a collection of algorithms, but as an enabling technology for solving real-world engineering problems within larger, multidisciplinary efforts.

4 Conclusion

This work illustrates how embedding computer graphics education within authentic engineering challenges can deepen student learning and broaden their perception of the field. Through interdisciplinary projects and carefully designed scaffolding, students connected theoretical graphics knowledge with real-world problem solving. This experiential approach reframes computer graphics as a platform for innovation and societal impact, beyond a subject confined to visual effects and entertainment.

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